

# Regiment Analysis

## A Multilevel Theory of Structure, Incentives, Behaviour, and Performance

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### Abstract

This paper develops Regiment Analysis as a general framework for studying hierarchical collections of observations after a Regiment-Sort structure has been identified or reconstructed. A regiment is treated as a mesoscopic unit: larger than an individual observation but smaller than the complete population or organization. The framework combines within-regiment statistics, between-regiment comparisons, multilevel models, incentives, rewards, punishments, behavioural responses, causal inference, and dynamic analysis. The resulting framework connects statistics, econometrics, organizational psychology, behavioural economics, finance, physics, chemistry, biology, medicine, political science, international relations, welfare economics, computer science, data science, and artificial intelligence. Particular attention is given to the distinction between performance and incentives, since high performance may cause rewards rather than result from them. Regiment Analysis therefore separates descriptive association, prediction, and causal identification. The paper proposes a compact computational architecture and identifies applications in firms, financial institutions, public administration, education, biological systems, scientific networks, and other hierarchical systems.

The paper ends with “The End”

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## 1 Introduction

Regiment Sort creates an ordered collection of internally sorted groups. Detection asks whether such a structure exists in observed data, while adversarial detection asks whether the observable structure identifies the mechanism that generated it. Regiment Analysis begins after these questions and asks a different question:

What can be learned from the internal structure, comparative performance, incentives, and dynamics of the recovered regiments?

This distinction is important. An array may possess a regiment structure without that structure having a useful interpretation. Conversely, once a meaningful grouping has been established, the groups may contain information about heterogeneity, organizational performance, incentives, risk, coordination, and behavioural response.

The idea is closely related to multilevel research. Organizations, for example, naturally contain individuals nested within teams, departments, firms, and wider institutional environments. Multilevel statistical methods were developed precisely to prevent phenomena operating at one level from being incorrectly attributed to another [1, 2, 3].

Regiment Analysis adopts this logic while beginning from a particular mathematical partition generated or detected through Regiment Sort.

## 2 The Regiment as a Mesoscopic Unit

Let an observed system contain  $N$  observations. Suppose the observations have been partitioned into  $r$  regiments, each containing  $q$  observations:

$$N = rq. \tag{1}$$

The  $i$ -th regiment is

$$R_i = (x_{i1}, x_{i2}, \dots, x_{iq}). \tag{2}$$

A regiment-level score is denoted by

$$S_i = G(R_i). \tag{3}$$

For the original Regiment-Sort construction, the regiments are ordered so that

$$S_1 \geq S_2 \geq \dots \geq S_r. \tag{4}$$

The regiment therefore occupies an intermediate level of analysis:

$$\text{individual observation} \longrightarrow \text{regiment} \longrightarrow \text{population or organization}. \tag{5}$$

This intermediate level is analogous to the team, department, portfolio, biological population, chemical cluster, or political bloc found in other disciplines.

### 3 Within-Regiment Analysis

For regiment  $i$ , define the mean

$$\bar{x}_i = \frac{1}{q} \sum_{j=1}^q x_{ij}. \quad (6)$$

Its variance is

$$s_i^2 = \frac{1}{q-1} \sum_{j=1}^q (x_{ij} - \bar{x}_i)^2. \quad (7)$$

Other useful quantities include the median, range, interquartile range, skewness, kurtosis, concentration, entropy, and selected order statistics.

The regiment profile can therefore be represented by

$$\Phi_i = (\bar{x}_i, s_i^2, \text{skew}_i, \text{kurt}_i, S_i, \dots). \quad (8)$$

The collection

$$\mathcal{P} = \{\Phi_1, \Phi_2, \dots, \Phi_r\} \quad (9)$$

constitutes the empirical Regiment Profile of the system.

### 4 Within- and Between-Regiment Variation

A central objective is to distinguish variation occurring inside regiments from variation occurring between regiments.

Let

$$\mu = \frac{1}{r} \sum_{i=1}^r \bar{x}_i. \quad (10)$$

A simple variance decomposition is

$$\text{Var}(X) = \text{Var}_{\text{between}}(X) + \text{Var}_{\text{within}}(X). \quad (11)$$

In balanced designs, the between-regiment component can be represented by

$$V_B = \frac{q}{N} \sum_{i=1}^r (\bar{x}_i - \mu)^2, \quad (12)$$

while the within-regiment component is

$$V_W = \frac{1}{N} \sum_{i=1}^r \sum_{j=1}^q (x_{ij} - \bar{x}_i)^2. \quad (13)$$

Thus,

$$V_T = V_B + V_W. \quad (14)$$

This decomposition connects Regiment Analysis to ANOVA, variance-components analysis, hierarchical linear models, and random-effects econometrics.

## 5 A Multilevel Statistical Model

A basic regiment-level model is

$$x_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad (15)$$

where  $\alpha_i$  is the regiment effect and  $\varepsilon_{ij}$  is the individual-level disturbance.  
A richer longitudinal specification is

$$x_{ijt} = \mu + \alpha_i + \lambda_t + \beta' Z_{ijt} + \varepsilon_{ijt}, \quad (16)$$

where  $Z_{ijt}$  contains observable covariates and  $\lambda_t$  represents common time effects.  
The regiment effect can itself be modelled as

$$\alpha_i = u_i + \gamma' W_i, \quad (17)$$

where  $W_i$  contains regiment-level characteristics.

This formulation is compatible with hierarchical linear modelling and longitudinal multilevel analysis [1, 4].

## 6 Rewards and Punishments

The principal extension developed here is to incorporate observable incentives.

For regiment  $i$ , define

$$I_i = (W_i, P_i), \quad (18)$$

where  $W_i$  represents rewards and  $P_i$  represents punishments or sanctions.

Examples of rewards include bonuses, promotions, awards, recognition, increased resources, improved assignments, and other measurable benefits. Punishments may include fines, warnings, demotions, disciplinary procedures, loss of privileges, reassignment, or termination.

The augmented regiment score becomes

$$S_i = G(R_i, W_i, P_i). \quad (19)$$

This extension makes Regiment Analysis substantially more applicable to administrative and organizational datasets.

## 7 Performance Is Not the Same as Incentives

A fundamental methodological principle is

$$\text{Performance} \neq \text{Reward} \neq \text{Punishment}. \quad (20)$$

A positive association between performance and rewards does not establish that rewards caused performance.

For example,

$$Y_i \longrightarrow W_i \quad (21)$$

is consistent with rewarding successful regiments.

But an incentive effect would instead suggest

$$W_i \longrightarrow Y_i. \quad (22)$$

Both processes may operate simultaneously:

$$W_{it} \longrightarrow Y_{i,t+1} \longrightarrow W_{i,t+1}. \quad (23)$$

Consequently, Regiment Analysis must distinguish correlation, prediction, and causal inference.

## 8 The Behavioural Response to Incentives

Let  $Y_{it}$  denote regiment performance at time  $t$ . A basic dynamic specification is

$$Y_{i,t+1} = \alpha_i + \rho Y_{it} + \beta W_{it} - \gamma P_{it} + \delta' X_{it} + \varepsilon_{i,t+1}. \quad (24)$$

Here  $\beta$  represents the conditional reward response and  $\gamma$  the conditional punishment response. The framework does not require either coefficient to be positive.

Rewards can produce unintended effects, including gaming, crowding out of intrinsic motivation, excessive risk-taking, threshold behaviour, or strategic manipulation. Punishments may similarly produce caution, concealment, withdrawal, or reduced experimentation. The behavioural-economics literature documents several mechanisms through which incentives can produce outcomes different from those intended by the designer [6].

Thus the empirical question is not simply whether incentives exist, but how different regiments respond to them.

## 9 Regiment-Specific Elasticities

For a continuous reward variable, an approximate reward elasticity can be defined as

$$\eta_i^W = \frac{\partial \ln Y_i}{\partial \ln W_i}. \quad (25)$$

Similarly,

$$\eta_i^P = -\frac{\partial \ln Y_i}{\partial \ln P_i}. \quad (26)$$

These quantities permit heterogeneous incentive responses.

One regiment may be highly reward-responsive, another sanction-responsive, and another relatively insensitive to either.

The result is a behavioural profile

$$B_i = (S_i, \eta_i^W, \eta_i^P, V_i, C_i, \dots), \quad (27)$$

where  $V_i$  may measure internal variance and  $C_i$  may measure cohesion.

## 10 Organizational Psychology

Organizational psychology provides a natural interpretation of the regiment as a group-level behavioural unit.

Individuals may be influenced by leadership, organizational climate, perceived fairness, group norms, social identification, autonomy, competence, workload, psychological safety, and incentives. Team-level states can emerge from individual perceptions and interactions and subsequently influence individual behaviour [5].

Regiment Analysis therefore permits a three-level interpretation:

$$\text{individual psychology} \rightarrow \text{regiment behaviour} \rightarrow \text{organizational outcome}. \quad (28)$$

The regiment need not correspond literally to a military formation. It may represent a work team, classroom, investment desk, hospital unit, administrative department, online community, laboratory group, or other empirically recoverable cluster.

## 11 Motivation and Incentives

A simple behavioural model is

$$E_{it} = F(A_{it}, M_{it}, W_{it}, P_{it}, C_{it}), \quad (29)$$

where  $E$  denotes effort,  $A$  ability,  $M$  motivation,  $W$  reward,  $P$  punishment, and  $C$  contextual conditions.

The framework deliberately avoids assuming that extrinsic incentives dominate intrinsic motivation. Contemporary psychological research emphasizes that motivation is affected by autonomy, competence, relatedness, internalization, and the design of external incentives [13].

Consequently, the same reward can produce different effects in different regiments.

## 12 Economics and Principal-Agent Theory

Regiment Analysis has an immediate interpretation in contract theory.

Suppose a principal chooses incentives for regiment  $i$ , while members of the regiment choose effort  $e_i$ . Let output be

$$Y_i = f(e_i, \theta_i) + \varepsilon_i, \quad (30)$$

where  $\theta_i$  represents latent productive conditions.

The principal chooses

$$(W_i, P_i) \quad (31)$$

to influence the agent's behaviour.

This is a standard principal-agent environment with heterogeneous groups. Incentive schemes may improve alignment, but imperfect observability and risk can generate unintended responses [7, 8].

A central implication is that observed performance should not automatically be interpreted as observed effort.

## 13 Risk and Incentive Design

Suppose

$$Y_i = \mu_i + \varepsilon_i, \quad (32)$$

with

$$\text{Var}(\varepsilon_i) = \sigma_i^2. \quad (33)$$

When performance is noisy, performance-contingent compensation may impose risk on agents.

Thus a simple incentive index should account for both reward sensitivity and outcome volatility:

$$\mathcal{I}_i = \frac{\partial W_i}{\partial Y_i} \frac{1}{\sigma_i}. \quad (34)$$

This connects Regiment Analysis with corporate finance, executive compensation, risk sharing, and financial economics [9, 10].

## 14 Finance

A financial portfolio can be treated as a regiment whose members are securities:

$$R_i = (r_{i1}, r_{i2}, \dots, r_{iq}). \quad (35)$$

The regiment score may be

$$S_i = \sum_{j=1}^q w_{ij} r_{ij}. \quad (36)$$

Regiment Analysis can then examine:

- return dispersion within portfolios;
- volatility differences between portfolios;
- concentration;

- drawdowns;
- factor exposure;
- incentive compensation;
- risk-adjusted performance;
- behavioural responses to gains and losses.

The same framework can therefore be applied to investment teams, trading desks, portfolios, or groups of financial instruments.

## 15 Econometrics

A panel model with regiment effects can be written as

$$Y_{it} = \alpha_i + \lambda_t + \beta W_{it} - \gamma P_{it} + \delta' X_{it} + \varepsilon_{it}. \quad (37)$$

Potential estimators include fixed effects, random effects, instrumental variables, difference-in-differences, event studies, regression discontinuity, randomized experiments, and dynamic panel estimators.

The appropriate estimator depends on the data-generating process rather than on the Regiment Analysis terminology.

A key identification problem is reverse causality:

$$Y_{it} \rightarrow W_{it}. \quad (38)$$

An instrument  $Z_{it}$  may be useful if

$$\text{Cov}(Z_{it}, W_{it}) \neq 0, \quad (39)$$

$$\text{Cov}(Z_{it}, \varepsilon_{it}) = 0. \quad (40)$$

Natural experiments and randomized incentive schemes are particularly valuable.

## 16 Thresholds and Strategic Behaviour

Suppose a reward is granted when

$$Y_i \geq c. \quad (41)$$

The observed distribution may then exhibit bunching around  $c$ .

Such behaviour is economically meaningful rather than merely statistical noise. Empirical work on performance incentives has found that workers may adjust behaviour around incentive thresholds and that incentive contracts can affect learning and productivity [11].

Regiment Analysis can therefore treat threshold effects as part of the regiment's behavioural signature.

## 17 The Ratchet Problem

Dynamic incentives may alter future targets.

Suppose

$$T_{i,t+1} = H(Y_{it}), \quad (42)$$

where  $T$  is the future performance target.

A rational regiment may then choose current output while anticipating future consequences.

This creates a possible ratchet effect:

$$Y_{it} \uparrow \longrightarrow T_{i,t+1} \uparrow \longrightarrow \text{future effort incentive} \downarrow. \quad (43)$$

The literature documents this phenomenon in performance-pay environments [12].

Thus a high-performing regiment need not remain high-performing when its incentive environment changes.



## 18 Welfare Economics

An incentive scheme should ultimately be evaluated by welfare rather than by the reward-induced output statistic alone.

Let social welfare be

$$\mathcal{W} = B(Y) - C(e) - D(P) + R(W), \quad (44)$$

where  $B$  represents benefits,  $C$  effort costs,  $D$  punishment costs, and  $R$  any broader social consequences of rewards.

The socially optimal incentive system satisfies

$$\max_{W,P} \mathcal{W}(W,P). \quad (45)$$

Private optimization and social optimization may differ.

A firm may increase measured productivity while reducing worker welfare. A government may improve compliance while increasing administrative burden. A school may improve a measured examination outcome while reducing unmeasured educational objectives.

Regiment Analysis should therefore distinguish objective performance from welfare.

## 19 Psychiatry and Behavioural Health

In clinical and behavioural datasets, a regiment may represent a treatment group, clinical team, patient cluster, or longitudinal subgroup.

Potential variables include:

$$Y_{it} = \text{clinical outcome}, \quad W_{it} = \text{positive reinforcement}, \quad P_{it} = \text{adverse consequence}. \quad (46)$$

The framework must not assume that punishment is therapeutically desirable. Rather, it provides a statistical language for studying observed incentive structures and their associations with outcomes.

Ethical safeguards, informed consent, privacy, and clinical validity remain prior constraints.

## 20 Biology

Biological systems frequently contain nested units.

Examples include:

$$\text{cell} \rightarrow \text{tissue} \rightarrow \text{organ} \rightarrow \text{organism} \rightarrow \text{population}. \quad (47)$$

A biological regiment may therefore be a cluster of cells, genes, organisms, or ecological units.

Regiment Analysis can study within-cluster heterogeneity, between-cluster variation, selection pressures, resource allocation, and adaptation.

The analogy should remain mathematical rather than implying that biological systems literally implement Regiment Sort.

## 21 Chemistry and Physics

In chemistry, a regiment may represent a molecular or reaction-state cluster. Within-regiment variation can describe conformational or energetic heterogeneity.

In statistical physics, the same structure resembles a coarse-grained ensemble:

$$\text{microstates} \rightarrow \text{mesoscopic clusters} \rightarrow \text{macroscopic state}. \quad (48)$$

The regiment therefore becomes a possible mesoscopic analytical unit.

This analogy is particularly useful because many physical systems are studied through aggregation from microscopic states into statistically meaningful collective states.

## 22 Political Science and Public Administration

Government departments, municipalities, legislative coalitions, administrative units, and policy teams can be represented as regiments.

Let

$$Y_{it} = \text{policy performance}, \quad (49)$$

with

$$W_{it} = \text{budgetary or political reward} \quad (50)$$

and

$$P_{it} = \text{political or administrative sanction}. \quad (51)$$

Regiment Analysis can then examine whether organizational structure predicts policy outcomes and whether incentives alter subsequent performance.

## 23 Geopolitics and International Relations

At the international level, a regiment may represent a bloc of states, alliance, economic coalition, diplomatic grouping, or regional organization.

The variables need not be military.

They may include:

- trade;
- investment;
- diplomatic alignment;
- development assistance;
- sanctions;
- institutional cooperation;
- voting behaviour;
- economic performance.

A diplomatic reward may increase cooperation, while a sanction may decrease cooperation. But causal interpretation remains difficult because states select into diplomatic relationships.

Regiment Analysis therefore provides a descriptive and statistical framework rather than a presumption about geopolitical causality.

## 24 Diplomacy

Diplomatic relationships can also be represented dynamically:

$$C_{ij,t+1} = F(C_{ij,t}, R_{ij,t}, P_{ij,t}, X_{ij,t}), \quad (52)$$

where  $C$  represents cooperation,  $R$  diplomatic inducements, and  $P$  diplomatic penalties.

This framework emphasizes feedback rather than static rankings.

A successful diplomatic relationship may generate rewards, which reinforce cooperation, while sanctions may alter behaviour but also produce countermeasures.

## 25 Warfare Economics and Abstract Conflict Systems

The framework can also describe abstract resource-allocation systems in which groups receive rewards for performance and sanctions for failure.

The mathematical object is

$$Y_{it+1} = F(Y_{it}, W_{it}, P_{it}, X_{it}). \quad (53)$$

The analysis concerns incentives and organizational performance, not operational battlefield tactics.

The same mathematical framework can therefore be applied to historical military administration, logistics, personnel systems, or defence economics without prescribing real-world violence.

## 26 Religion and Philosophy

Religious institutions provide another class of hierarchical organizations in which communities, congregations, schools, orders, or charitable organizations can constitute mesoscopic units.

Rewards and punishments may be material, social, institutional, or symbolic.

Philosophically, Regiment Analysis raises a classical question:

Does behaviour reveal the individual, the institution, the incentive structure, or the interaction among them?

The answer is generally not identifiable from a single observational level.

This resembles the broader philosophical problem of emergence: properties of a group may not be reducible to properties of its members considered independently.

## 27 Logic and Causal Structure

The distinction between association and causation can be represented logically.

The statement

$$W_i \Rightarrow Y_i \quad (54)$$

is a causal hypothesis.

The empirical observation

$$\text{Cov}(W_i, Y_i) > 0 \quad (55)$$

is merely an association.

The latter does not logically entail the former.

Possible causal structures include

$$W_i \rightarrow Y_i, \quad (56)$$

$$Y_i \rightarrow W_i, \quad (57)$$

$$W_i \leftarrow Z_i \rightarrow Y_i, \quad (58)$$

$$W_i \rightarrow Y_i \rightarrow W_{i,t+1}. \quad (59)$$

Regiment Analysis therefore requires explicit causal assumptions when causal claims are made.

## 28 Data Science Representation

Each regiment can be represented as a row in a higher-level dataset:

$$D_R = \{(\Phi_i, I_i, B_i, Y_i)\}_{i=1}^r. \quad (60)$$

This produces a two-stage data architecture:

$$D_{\text{raw}} \rightarrow D_{\text{regiment}} \rightarrow D_{\text{model}}. \quad (61)$$

The first transformation aggregates observations into regiments. The second analyzes regiment-level features.

This architecture is useful for large datasets because computation can occur independently within regiments before higher-level analysis.

## 29 Machine Learning

Regiment profiles can be used as machine-learning features:

$$\Phi_i \longrightarrow \hat{Y}_i. \quad (62)$$

Possible methods include:

- clustering;
- principal component analysis;
- random forests;
- gradient boosting;
- support-vector machines;
- neural networks;
- representation learning;
- graph neural networks;
- temporal models.

A particularly natural application is anomaly detection.

Let

$$A_i = d(\Phi_i, \hat{\Phi}_i), \quad (63)$$

where  $d$  measures deviation from the expected regiment profile.

Large  $A_i$  identifies a potentially anomalous regiment.

## 30 Artificial Intelligence and Adaptive Systems

An adaptive system can use Regiment Analysis in a feedback loop:

$$\text{data} \rightarrow \text{regiment detection} \rightarrow \text{regiment analysis} \rightarrow \text{prediction} \rightarrow \text{intervention} \rightarrow \text{new data}. \quad (64)$$

This creates a potential policy-learning architecture.

However, intervention changes the data-generating process. Consequently, a predictive model trained on passive data may cease to be valid after deployment.

This is a fundamental distinction between prediction and control.

## 31 A General Behavioural Feedback System

The complete system can be represented as

$$R_{i,t} \rightarrow Y_{i,t}, \quad (65)$$

$$Y_{i,t} \rightarrow (W_{i,t}, P_{i,t}), \quad (66)$$

$$(W_{i,t}, P_{i,t}) \rightarrow Y_{i,t+1}. \quad (67)$$

Thus,

Structure  $\rightarrow$  Performance  $\rightarrow$  Evaluation  $\rightarrow$  Reward/Punishment  $\rightarrow$  Behaviour  $\rightarrow$  Future Performance

(68)

This feedback loop is the central behavioural extension of Regiment Analysis.

## 32 Computational Architecture

Figure 1 summarizes the proposed architecture.

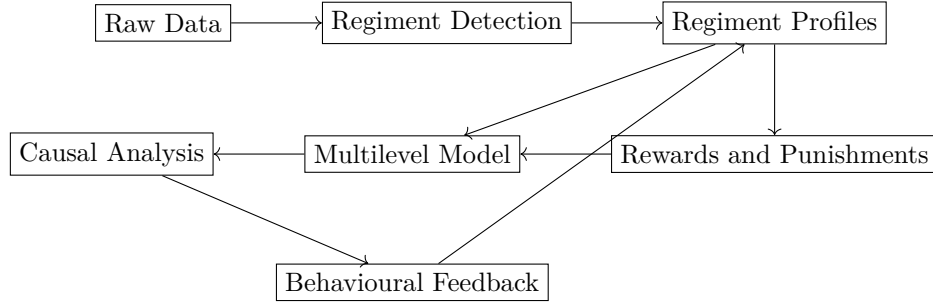


Figure 1: A general architecture for Regiment Analysis.

## 33 Algorithm

A simple procedure is:

**Input:** observed data  $D$ , candidate regiment structure, performance variables, rewards, punishments.

**Step 1:** Detect or reconstruct candidate regiments.

**Step 2:** Calculate within-regiment statistics.

**Step 3:** Calculate between-regiment statistics.

**Step 4:** Construct regiment profiles.

**Step 5:** Merge performance, reward, punishment, and contextual variables.

**Step 6:** Estimate descriptive and multilevel models.

**Step 7:** Test for reverse causality and confounding.

**Step 8:** Where credible identification exists, estimate causal effects.

**Step 9:** Examine dynamic and heterogeneous responses.

**Step 10:** Validate the model on held-out observations or subsequent periods.

A compact pseudocode representation is:

REGIMENT\_ANALYSIS( $D, q$ ):

```

R <- DETECT_REGIMENTS(D, q)

for each regiment i in R:
  profile[i] <- COMPUTE_PROFILE(R[i])
  incentives[i] <- OBSERVE_INCENTIVES(R[i])
  performance[i] <- OBSERVE_PERFORMANCE(R[i])

panel <- MERGE(profile, incentives, performance)

descriptive <- DESCRIPTIVE_ANALYSIS(panel)
multilevel <- MULTILEVEL_MODEL(panel)
  
```

```

causal <- CAUSAL_IDENTIFICATION(panel)
dynamic <- DYNAMIC_ANALYSIS(panel)
heterogeneity <- REGIMENT_HETEROGENEITY(panel)

return descriptive, multilevel, causal,
        dynamic, heterogeneity

```

## 34 Statistical Diagnostics

A useful empirical dashboard is shown in Table 1.

Table 1: Core Regiment Analysis diagnostics

| Dimension     | Statistic            | Interpretation               |
|---------------|----------------------|------------------------------|
| Location      | Mean/median          | Central performance          |
| Dispersion    | Variance/IQR         | Internal heterogeneity       |
| Shape         | Skewness/kurtosis    | Distributional structure     |
| Concentration | Gini/HHI             | Internal concentration       |
| Cohesion      | Correlation          | Internal dependence          |
| Performance   | $S_i$                | Regiment aggregate           |
| Reward        | $W_i$                | Positive incentive intensity |
| Punishment    | $P_i$                | Sanction intensity           |
| Response      | $\eta_i^W, \eta_i^P$ | Incentive sensitivity        |
| Persistence   | Autocorrelation      | Temporal stability           |
| Anomaly       | $A_i$                | Structural deviation         |

## 35 Propositions

### 35.1 Proposition 1: Hierarchical Decomposition

If the observations are partitioned into disjoint regiments, total variation can be decomposed into within-regiment and between-regiment components.

**Proof.** For every observation,

$$x_{ij} - \mu = (x_{ij} - \bar{x}_i) + (\bar{x}_i - \mu). \quad (69)$$

Squaring and summing over all observations yields the standard within-between decomposition because the cross-product term vanishes within each regiment.  $\square$

### 35.2 Proposition 2: Association Does Not Establish Incentive Causality

If rewards are assigned as a function of contemporaneous performance, then a positive covariance between rewards and performance does not establish a positive causal effect of rewards on future performance.

**Proof.** Suppose

$$W_i = h(Y_i, \nu_i). \quad (70)$$

Then  $W_i$  may be correlated with  $Y_i$  even when

$$\frac{\partial Y_i}{\partial W_i} = 0. \quad (71)$$

Therefore observational covariance alone does not identify the causal derivative.  $\square$

### 35.3 Proposition 3: Heterogeneous Incentive Responses

Suppose

$$Y_{i,t+1} = \alpha_i + \beta_i W_{it} + \varepsilon_{i,t+1}. \quad (72)$$

If  $\beta_i$  varies across regiments, a homogeneous incentive coefficient estimates an average response rather than the complete behavioural structure.

**Proof.** The homogeneous model imposes

$$\beta_i = \beta \quad (73)$$

for every  $i$ . If this restriction is false, the estimated coefficient aggregates heterogeneous responses and cannot represent every regiment simultaneously.  $\square$

## 36 A General Regiment Utility Function

A broad economic representation is

$$U_i = B(Y_i) - C(e_i) + \omega W_i - \pi P_i - \Omega_i, \quad (74)$$

where  $B$  is benefit,  $C$  effort cost,  $\omega$  the value of reward,  $\pi$  the cost of punishment, and  $\Omega_i$  other organizational costs.

The behavioural problem is

$$\max_{e_i} U_i. \quad (75)$$

The organizational problem is then

$$\max_{W_i, P_i} \mathcal{W}, \quad (76)$$

subject to the behavioural responses of the regiments.

This formulation connects Regiment Analysis to mechanism design, contract theory, personnel economics, behavioural economics, and welfare economics.

## 37 Regiment Competition

When several regiments compete for a common reward pool, incentives become relative.

Let

$$W_i = f(S_i - S_{-i}), \quad (77)$$

where  $S_{-i}$  represents the performance of competing regiments.

Relative-performance incentives can produce cooperation, competition, information sharing, sabotage, or strategic withholding depending on the institutional environment.

The analysis therefore requires attention to the game played among regiments rather than only the relationship between one regiment and its principal.

## 38 Networks and Inter-Regiment Dependence

Regiments may not be statistically independent.

Let  $A$  be an inter-regiment adjacency matrix. Then a network model may be written as

$$Y_t = \rho A Y_t + X_t \beta + \varepsilon_t. \quad (78)$$

This permits Regiment Analysis to incorporate supply chains, communication networks, financial exposures, political alliances, scientific collaborations, and social relationships.

A regiment can therefore be both a statistical cluster and a node in a larger network.

## 39 Robustness and Adversarial Considerations

The preceding paper on adversarial detection established that an observable Regiment-Sort structure may be deliberately mimicked.

Regiment Analysis must therefore distinguish:

$$\text{observed regiment} \neq \text{known generating mechanism.} \quad (79)$$

Robust analysis should use multiple data sources where possible, including independent administrative records, timestamps, external measurements, interventions, and provenance information.

This is especially important when incentives themselves can be manipulated.

## 40 Missing Data and Sentinel Values

Real-world datasets frequently contain missing observations.

A structural missingness indicator  $M_{ij}$  can be defined by

$$M_{ij} = \begin{cases} 1, & \text{if observation is missing,} \\ 0, & \text{otherwise.} \end{cases} \quad (80)$$

Missing observations should not automatically be treated as zero.

Where a Regiment-Sort implementation uses a padding sentinel, the sentinel should remain outside substantive statistics:

$$S_i = \sum_{j: M_{ij}=0} x_{ij}. \quad (81)$$

This preserves the distinction between structural padding and economically or scientifically meaningful observations.

## 41 Comparison Across Systems

Suppose two systems contain regiment profiles

$$\mathcal{P}^A = \{\Phi_1^A, \dots, \Phi_r^A\} \quad (82)$$

and

$$\mathcal{P}^B = \{\Phi_1^B, \dots, \Phi_r^B\}. \quad (83)$$

A distance between systems may be defined as

$$d(\mathcal{P}^A, \mathcal{P}^B) = \min_{\pi} \sum_{i=1}^r d_0(\Phi_i^A, \Phi_{\pi(i)}^B), \quad (84)$$

where  $\pi$  ranges over admissible permutations.

This permits comparative organizational analysis even when individual observations are not directly matched.



## 42 Applications Across Fields

Table 2: Illustrative applications of Regiment Analysis

| Field                     | Regiment                | Possible outcome           |
|---------------------------|-------------------------|----------------------------|
| Economics                 | Worker/team             | Productivity               |
| Finance                   | Portfolio/desk          | Return and risk            |
| Organizational psychology | Work group              | Engagement and performance |
| Psychology                | Behavioural subgroup    | Response to intervention   |
| Psychiatry                | Clinical group          | Treatment outcome          |
| Biology                   | Population/cell cluster | Growth or adaptation       |
| Chemistry                 | Molecular cluster       | Reaction or energy state   |
| Physics                   | Mesoscopic ensemble     | Collective state           |
| Education                 | Classroom/school group  | Learning outcome           |
| Political science         | Policy bloc             | Policy performance         |
| International relations   | State coalition         | Cooperation                |
| Diplomacy                 | Diplomatic cluster      | Relationship stability     |
| Public administration     | Department              | Service delivery           |
| Welfare economics         | Population subgroup     | Social welfare             |
| Computer science          | Data partition          | Algorithmic performance    |
| AI/ML                     | Latent cluster          | Prediction or anomaly      |

## 43 The General Theory

The preceding constructions suggest that Regiment Analysis can be summarized by four layers.

$$\boxed{\text{Structure} \rightarrow \text{Profile} \rightarrow \text{Incentive} \rightarrow \text{Response}} \quad (85)$$

The first layer identifies the regiment.

The second describes its internal and external characteristics.

The third records the incentive environment.

The fourth estimates how the regiment responds.

For longitudinal systems this becomes

$$\boxed{R_{i,t} \rightarrow Y_{i,t} \rightarrow (W_{i,t}, P_{i,t}) \rightarrow Y_{i,t+1} \rightarrow R_{i,t+1}} \quad (86)$$

The final arrow emphasizes that interventions can alter the future structure itself.

## 44 Research Programme

The framework suggests several empirical research programmes.

First, one may estimate the distribution of regiment effects:

$$\alpha_i \sim F_\alpha. \quad (87)$$

Second, one may estimate the distribution of incentive responses:

$$\beta_i \sim F_\beta. \quad (88)$$

Third, one may investigate whether structural characteristics predict behavioural responsiveness:

$$\beta_i = \theta' \Phi_i + \nu_i. \quad (89)$$

Fourth, one may examine whether incentive responses alter future regiment composition:

$$\Pr(R_{i,t+1} = k) = F(R_{it}, Y_{it}, W_{it}, P_{it}, X_{it}). \quad (90)$$

This turns Regiment Analysis into a potentially dynamic theory of organizational evolution.

## 45 Limitations

Regiment Analysis does not by itself solve causal identification.

Nor does the discovery of a statistically coherent regiment establish that the regiment corresponds to a natural organizational unit.

Three distinctions must therefore remain explicit:

$$\text{statistical structure} \neq \text{organizational reality}, \quad (91)$$

$$\text{organizational association} \neq \text{causal mechanism}, \quad (92)$$

$$\text{causal effect} \neq \text{social welfare}. \quad (93)$$

The framework is consequently best viewed as a structured analytical language rather than a universal estimator.

## 46 Conclusion

Regiment Analysis extends the Regiment-Sort programme from structural detection to substantive interpretation.

A regiment is treated as a mesoscopic unit whose members possess individual characteristics but whose collective behaviour may contain additional information. This makes it possible to study within-regiment heterogeneity, between-regiment inequality, organizational effects, incentives, rewards, punishments, behavioural responses, dynamic feedback, networks, and causal identification.

The central empirical principle is simple:

*Do not merely ask which regiment performs better.  
Ask why, under what incentives, with what internal structure,  
and with what consequences.*

(94)

The resulting progression is

Regiment Sort  $\rightarrow$  Regiment Detection  $\rightarrow$  Adversarial Detection  $\rightarrow$  Regiment Analysis.

(95)

The first creates structure. The second discovers structure. The third tests whether structure identifies mechanism. The fourth studies the structure itself.

In this sense, Regiment Analysis transforms a sorting procedure into a general multilevel empirical framework.

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## Glossary

### **Regiment**

A finite group of observations treated as a mesoscopic analytical unit.

### **Regiment Score**

An aggregate statistic used to characterize or rank a regiment.

### **Regiment Profile**

A vector of descriptive, behavioural, incentive, and performance characteristics associated with a regiment.

### **Within-Regiment Variation**

Variation among observations belonging to the same regiment.

### **Between-Regiment Variation**

Variation between regiment-level means or other regiment-level characteristics.

### **Reward**

An observable positive incentive associated with behaviour or performance.

### **Punishment**

An observable sanction or negative incentive associated with behaviour or performance.

### **Incentive Response**

The behavioural response of a regiment to a change in rewards or punishments.

### **Mesoscopic Unit**

An analytical unit between the individual observation and the complete population or organization.

### **Multilevel Model**

A statistical model allowing variables and effects to operate at multiple nested levels.

### **Regiment Effect**

A systematic component of an outcome associated with membership in a particular regiment.

### **Behavioural Feedback**

A process in which outcomes affect incentives that subsequently influence future outcomes.

### **Regiment Heterogeneity**

Differences in characteristics or behavioural responses across regiments.

**Regiment Analysis**

The systematic statistical, econometric, behavioural, and computational analysis of identified regiments.

**The End**